

KINDERSLEY, SK, Canada

WMO: 711290

Lat: 51.520N

Lon: 109.180W

Elev: 694

StdP: 93.26

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-29.2	-36.5	0.1	-32.2	-33.4	0.2	-29.1	13.5	-7.0	11.9	-9.4	3.0	190	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	30.7	17.5	28.8	17.1	26.8	16.4	19.6	26.9	18.5	25.6	17.5	24.4	5.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.4	22.4	16.0	12.3	21.1	14.9	11.5	20.0	59.1	27.2	55.3	25.7	51.9	24.4	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.4	9.2	DB	-35.9	34.4	2.9	1.8	-38.0	35.7	-39.7	36.8	-41.3	37.8	-43.4	39.1	(4)
(5)			WB	-36.0	21.7	2.9	1.5	-38.1	22.8	-39.7	23.7	-41.3	24.6	-43.4	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-13.3	-12.0	-4.9	4.1	10.9	15.4	18.4	17.6	12.4	4.4	-4.6	-11.5	(6)
(7)		DBStd	12.98	8.55	7.72	8.08	5.57	4.59	3.52	3.14	3.60	4.78	5.24	6.89	7.95	(7)
(8)		HDD10.0	3352	723	617	463	186	46	3	0	1	28	183	437	667	(8)
(9)		HDD18.3	5638	982	850	721	426	234	101	38	58	186	431	687	925	(9)
(10)		CDD10.0	852	0	0	0	9	72	165	260	236	99	11	0	0	(10)
(11)		CDD18.3	96	0	0	0	0	2	12	39	34	7	0	0	0	(11)
(12)		CDH23.3	1467	0	0	0	6	83	184	507	514	168	4	0	0	(12)
(13)		CDH26.7	446	0	0	0	1	16	44	157	177	53	0	0	0	(13)
(14)	Wind	WSAvg	4.4	4.1	4.1	4.7	5.0	5.0	4.6	4.0	3.9	4.2	4.5	4.3	4.1	(14)
(15)	Precipitation	PrecAvg	352	11	8	12	23	35	81	63	51	29	16	13	10	(15)
(16)		PrecMax	634	36	17	30	62	83	137	183	132	109	48	32	30	(16)
(17)		PrecMin	138	0	0	0	2	0	24	17	1	4	5	0	0	(17)
(18)		PrecStd	97	8	5	8	16	24	31	36	27	24	11	9	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.0	14.4	23.6	28.9	30.7	33.2	33.3	31.4	23.3	13.6	6.4	(19)
(20)			MCWB	2.3	1.7	7.9	11.4	14.2	17.0	19.5	17.7	16.5	12.6	7.5	3.0	(20)
(21)		2%	DB	2.7	2.3	10.5	19.8	25.7	27.6	30.4	30.8	27.9	18.8	9.3	3.6	(21)
(22)			MCWB	0.9	0.6	5.5	10.0	13.3	16.1	18.7	17.0	15.0	10.4	5.1	1.4	(22)
(23)		5%	DB	0.9	0.7	7.2	16.7	23.1	25.4	28.3	28.5	24.8	15.5	6.2	1.7	(23)
(24)			MCWB	-0.5	-0.6	3.7	8.0	12.4	15.3	18.1	17.0	13.9	8.9	3.0	0.0	(24)
(25)		10%	DB	-1.5	-1.2	4.4	13.8	20.5	23.2	26.2	26.2	21.8	12.8	3.8	-0.9	(25)
(26)			MCWB	-2.5	-2.2	2.0	6.6	11.0	14.3	17.3	16.3	12.6	7.4	1.6	-2.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.1	8.6	12.3	15.6	19.2	21.7	20.0	17.5	13.4	8.1	3.4	(27)
(28)			MCDWB	4.4	3.6	13.7	21.7	25.8	27.1	28.9	28.1	28.6	21.7	12.8	5.9	(28)
(29)		2%	WB	1.1	0.7	5.8	10.3	14.1	17.6	20.3	18.9	15.7	11.0	5.3	1.5	(29)
(30)			MCDWB	2.6	2.1	9.9	18.8	24.0	24.5	27.6	26.9	25.5	17.2	8.9	3.4	(30)
(31)		5%	WB	-0.5	-0.5	3.9	8.6	12.9	16.4	19.2	17.9	14.3	9.2	3.4	-0.1	(31)
(32)			MCDWB	0.9	0.7	6.9	15.5	21.4	23.1	25.9	25.6	23.3	14.8	5.8	1.7	(32)
(33)		10%	WB	-2.5	-2.2	2.1	7.2	11.7	15.3	18.1	16.9	13.1	7.7	1.7	-2.0	(33)
(34)			MCDWB	-1.5	-1.2	4.2	12.9	19.3	21.3	24.5	24.3	20.9	12.4	3.6	-0.9	(34)
(35)	Mean Daily Temperature Range	MDBR	9.5	9.9	9.7	12.4	13.4	11.7	12.9	13.6	13.5	11.9	9.3	9.2	(35)	
(36)		5% DB	MCDBR	10.6	10.4	12.8	17.6	17.5	15.4	16.3	17.9	19.0	16.7	12.5	11.0	(36)
(37)			MCWBR	9.3	9.1	8.6	9.4	8.2	6.8	7.1	7.3	8.5	9.5	8.7	9.1	(37)
(38)		5% WB	MCDWB	10.7	10.4	12.3	16.2	15.3	13.1	14.2	14.7	17.3	15.8	11.5	11.0	(38)
(39)			MCWBR	9.6	9.2	8.5	9.1	7.5	6.4	7.0	6.7	8.1	9.3	8.4	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.261	0.269	0.299	0.358	0.382	0.381	0.379	0.393	0.340	0.293	0.265	0.247	(40)	
(41)		taud	2.307	2.356	2.410	2.268	2.247	2.309	2.345	2.262	2.392	2.505	2.455	2.364	(41)	
(42)		Ebn at Noon	764	870	906	879	870	873	869	832	846	835	768	731	(42)	
(43)		Edh at Noon	63	79	92	120	129	123	117	120	94	69	54	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.17	2.22	3.58	4.83	5.87	6.04	6.53	5.27	3.81	2.29	1.18	0.84	(44)	
(45)		RadStd	0.14	0.19	0.29	0.42	0.41	0.47	0.29	0.42	0.35	0.30	0.12	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page